

Engineering Physics (PH1001-1)

MCQ Questions

Unit-II Semiconductors and Superconductors

1. Solids with high value of conductivity are called:

- (a) Conductors
- (b) Non-metal
- (c) Insulator
- (d) Semi-conductor

Ans: (a)

2. Fermi level for a metal is

- (a) Highest energy level occupied by electrons at 0°C
- (b) Average value of all available energy levels
- (c) Highest energy level occupied by electrons at 0 K
- (d) Addition of energy of all available electron energy levels

Ans: (c)

3. The probability of occupation of the electrons at any temperature is given as,

(a) $f(E) = \frac{1}{e^{(E+E_f)/k_B T} + 1}$

(b) $f(E) = \frac{1}{e^{(E-E_f)/k_B T} - 1}$

(c) $f(E) = \frac{1}{e^{(E-E_f)/k_B T} + 1}$

(d) (iv) $f(E) = \frac{1}{e^{(E-E_f)/k_B} + 1}$

Ans: (c)

4. The relationship between current density J and electric field E is

- (a) $J = \sigma E$
- (b) $J = \sigma / E$
- (c) $J = \sigma/2E$
- (d) $J = 1/ \sigma E$

Ans: (a)

5. Intrinsic semiconductors are those

- (a) Which are made of semiconductor material in its purest form
- (b) Which have zero energy gap
- (c) Which have more electrons than holes
- (d) Which are available locally

Ans: (a)

6. A pure semiconductor behaves like an insulator at 0^0 K because

- (a) There is no recombination of electrons with holes
- (b) Drift velocity of free electrons is very small
- (c) Free electrons are not available for current conduction
- (d) Energy possessed by electrons at that low temperature is almost zero

Ans: (c)

7. Which of the following about Fermi-Dirac distribution is false?

- (a) When $E = E_F$, the probability of finding an electron with energy equal to the Fermi energy in a metal is $\frac{1}{2}$ at any temperature.
- (b) At $T = 0$ K all the energy level up to E_F are occupied and all the energy levels above E_F are empty.
- (c) When $T > 0$ K, some levels above E_F are partially filled while some levels below E_F are partially empty.
- (d) When $T = 0$ K, some levels above E_F are partially filled while some levels below E_F are empty.

Ans: (d)

8. Examples of Fermions are

- (a) Electrons
- (b) Photons
- (c) Phonons
- (d) Atoms

Ans: (a)

9. Using Fermi distribution function, the value of $f(E)$ for $(E - E_F) = 0.01$ eV at 200 K is

- (a) 0.36
- (b) 0.64
- (c) 0.45
- (d) 0.55

Ans: (a)

10. The magnetic lines of force cannot penetrate the body of a superconductor, a phenomenon is known as

- (a) Isotopic effect
- (b) BCS theory
- (c) Meissner effect
- (d) Silsbee's effect

Ans: (c)

11. The minimum amount of current passed through the body of superconductor in order to destroy the superconductivity is called

- (a) Induced current
- (b) Critical current
- (c) Eddy current
- (d) Hall current

Ans: (b)

12. In superconductivity the conductivity of a material becomes

- (a) Zero
- (b) Finite
- (c) Infinite
- (d) None of the answers

Ans: (c)

13. In superconductors the temperature at which conductivity of a material becomes infinite is called

- (a) Critical temperature
- (b) Absolute temperature
- (c) Mean temperature
- (d) Crystallization temperature

Ans: (a)

14. The superconducting state is perfectly _____ in nature.

- (a) Diamagnetic
- (b) Paramagnetic
- (c) Ferromagnetic
- (d) Ferrimagnetic

Ans: (a)

15. The binding energy of a Cooper pair is of the order of _____

- (a) 10^{-3} eV
- (b) 10^3 eV
- (c) 10^{-3} J
- (d) 10^3 J

Ans: (a)

16. The electron pairs in a superconductor are _____

- (a) Bosons
- (b) Leptons
- (c) Hydrons
- (d) Fermions

Ans: (a)

17. The transition to normal state occurs abruptly at a critical magnetic field (H_c) in

- (a) Type-I superconductor
- (b) Type-II superconductor
- (c) Both Type-I and Type-II superconductors
- (d) Conductors

Ans: (a)

18. The magnetic susceptibility (χ) in superconductor is

- (a) positive
- (b) zero
- (c) negative
- (d) infinity

Ans: (c)

19. Hard superconductors are also called as _____

- (a) Type-I superconductor
- (b) Type-II superconductor
- (c) Both Type-I and Type-II superconductors
- (d) Conductors

Ans: (b)

20. Silsbee's rule gives the relation between

- (a) current and critical temperature
- (b) current and critical magnetic field
- (c) current and isotopic mass
- (d) critical current and critical magnetic field.

Ans: (d)

21. The critical temperature and critical magnetic field for superconducting lead are 7.2 K and 800 Gauss respectively. What will be the temperature up to which lead will be in superconducting state in a magnetic field of 400 gauss?

- (a) 1.5 K
- (b) 0.5 K
- (c) 15 K
- (d) 5.09 K

Ans: (d)

22. Superconducting tin has a critical magnetic field of 217 gauss at 2 K. If the critical temperature for superconducting transition for tin is 3.7 K, find the critical magnetic field at 3 K.

- (a) 105 Gauss
- (b) 10.5 Gauss
- (c) 1.05 Gauss
- (d) 150 Gauss

Ans: (a)

23. A semiconductor is formed by bonds.

- (a) Covalent
- (b) Electrovalent
- (c) Co-ordinate
- (d) Ionic

Ans: (a)

24. A semiconductor has temperature coefficient of resistance.

- (a) Positive
- (b) Zero
- (c) Negative
- (d) Infinite

Ans : (c)

25. The most commonly used semiconductor is

- (a) Gallium
- (b) Silicon
- (c) Carbon
- (d) Arsenide

Ans : (b)

26. A semiconductor has generally valence electrons.

- (a) 2
- (b) 3
- (c) 6
- (d) 4

Ans : (d)

27. When a pentavalent impurity is added to a pure semiconductor, it becomes

- (a) An insulator
- (b) An intrinsic semiconductor
- (c) p-type semiconductor
- (d) n-type semiconductor

Ans : (d)

28. Addition of pentavalent impurity to a semiconductor creates

- (a) Donor electrons
- (b) Holes
- (c) Valence electrons
- (d) Bound electrons

Ans : (a)

29. A pentavalent impurity has valence electrons

- (a) 3
- (b) 5
- (c) 4
- (d) 6

Ans : (b)

30. An n-type semiconductor is

- (a) Positively charged
- (b) Negatively charged
- (c) Electrically neutral
- (d) None of the answers

Ans : (c)

31. A trivalent impurity has valence electrons

- (a) 4
- (b) 5
- (c) 6
- (d) 3

Ans : (d)

32. Addition of trivalent impurity to a semiconductor creates

- (a) Holes
- (b) Donor electrons
- (c) Valence electrons
- (d) Bound electrons

Ans : (a)

33. A hole in a semiconductor is defined as

- (a) A free electron
- (b) Electron vacancy
- (c) A free proton
- (d) A free neutron

Ans : (b)

34. As the doping to a pure semiconductor increases, the bulk resistance of the semiconductor

- (a) Remains the same
- (b) Increases
- (c) Decreases
- (d) Becomes zero

Ans : (c)

35. A hole and electron in close proximity would tend to

- (a) Repel each other
- (b) Attract each other
- (c) Have no effect on each other
- (d) None of the answers

Ans : (b)

36. In an intrinsic semiconductor, current conduction is due to

- (a) Only holes
- (b) Only electrons
- (c) Both holes and electrons
- (d) None of the answers

Ans : (c)

37. When a pure semiconductor is heated, its resistance

- (a) Increases
- (b) Decreases
- (c) Remains the same
- (d) Can't say

Ans : (b)

38. In an intrinsic semiconductor, the number of free electrons

- (a) Equals the number of holes
- (b) Is greater than the number of holes
- (c) Is less than the number of holes
- (d) None of the answers

Ans (a)

39. At room temperature, an intrinsic semiconductor has

- (a) Holes only
- (b) Electrons and holes
- (c) Electrons only
- (d) No holes and no electrons

Ans : (b)

40. At absolute temperature, an intrinsic semiconductor has

- (a) A few free electrons
- (b) Many holes
- (c) Many free electrons
- (d) No holes and no free electrons

Ans : (d)

41. Which of the following is known as indirect band gap semiconductors?

- (a) Germanium
- (b) GaAs
- (c) GaAsP
- (d) Carbon

Ans : (a)

42. Which of the following is a semiconductor

- (a) Diamond
- (b) Arsenic
- (c) Phosphorous
- (d) Gallium arsenide

Ans : (d)

43. In an intrinsic semiconductor, the Fermi level

- (a) Lies at the middle of the forbidden energy gap.
- (b) Is near the conduction band.
- (c) Is near the valence band.
- (d) May be anywhere in the forbidden energy gap.

Ans : (a)

44. For silicon, the energy gap at 300 K is

- (a) 0.7 J
- (b) 1.1 J
- (c) 1.1 eV
- (d) 0.7 eV

Ans : (c)

45. The forbidden gap for germanium is,

- (a) 0.7 J
- (b) 0.7 eV
- (c) 1.1 eV
- (d) 1.1 J

Ans: (b)

46. In a N-type semiconductor, the position of Fermi-level

- (a) Is below the middle of the energy gap
- (b) Is at the middle of the energy gap
- (c) Is above the middle of the energy gap
- (d) Can be any where

Ans : (c)

47. The mobility of electrons in a material is expressed in unit of:

- (a) V/s
- (b) $\text{m}^2/\text{V}\cdot\text{s}$
- (c) m^2/s
- (d) J/K

Ans : (b)

48. The energy gap in a semiconductor

- (a) Increases with temperature
- (b) Does not change with temperature
- (c) Decreases with temperature
- (d) Is zero

Ans : (b)

49. Donor impurity atoms in semiconducting material results a new

- (a) Wide energy band
- (b) Narrow energy band
- (c) Discrete energy level just below conduction band
- (d) Discrete energy level just above valance band

Ans : (c)

50. Hall Effect is clearly visible in _____

- (a) Insulators
- (b) Semiconductors
- (c) Super conductors
- (d) Non metals

Ans : (b)

51. Which of the following represents correct expression for Lorentz force?

- (a) BeV
- (b) BV
- (c) eV
- (d) B

Ans : (a)

52. Hall effect can be used to measure

- (a) Magnetic field intensity
- (b) Mobility
- (c) Carrier concentration
- (d) All the answers

Ans : (d)

53. Which of the following parameters can't be found with Hall Effect?

- (a) Polarity
- (b) Conductivity
- (c) Carrier concentration
- (d) Area of the device

Ans : (d)

54. In the Hall Effect, the electric field is in X direction and the velocity is in Y direction. What is the direction of the magnetic field?

- (a) X
- (b) Y
- (c) Z
- (d) XY plane

Ans : (c)

55. The number of electrons in a semiconductor is 10^{20} . Then the Hall coefficient is

- (a) 0.625
- (b) 0.0625
- (c) 6.25
- (d) 62.5

Ans : (b)

56. Calculate the conductivity of silicon doped with 10^{21} atoms m^{-3} of boron if the mobility of holes is $0.048 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$.

- (a) $76.8/\Omega\text{m}$
- (b) $7.68/\Omega\text{m}$
- (c) $7.68 \Omega\text{m}$
- (d) $0.768/\Omega\text{m}$

Ans : (b)

57. Calculate the resistivity of intrinsic germanium if the intrinsic carrier density is $2.5 \times 10^{19} \text{ m}^{-3}$ assuming electron and hole mobilities of 0.38 and $0.18 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$ respectively.

- (a) $0.45/\Omega\text{m}$
- (b) $0.045 \Omega\text{m}$
- (c) $0.45 \Omega\text{m}$
- (d) $4.50 \Omega\text{m}$

Ans : (c)

58. A semiconductor sample of thickness $1.2 \times 10^{-4} \text{ m}$ is placed in a magnetic field of 0.2 T acting perpendicular to its thickness. The Hall voltage generated when a current of 100 mA passes through it is (Assume the carrier concentration to be 10^{23} m^{-3})

- (a) 0.123 V
- (b) 0.0123 V
- (c) 1.23 V
- (d) 0.0012 V

Ans : (b)

59. Intrinsic silicon has a carrier concentration of $1.1 \times 10^{16} \text{ m}^{-3}$. If the mobilities of electrons and holes are 0.17 and $0.035 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ respectively at room temperature, the resistivity of silicon is

- (a) $0.277 \times 10^3 \Omega \text{ m}$
- (b) $27.7 \times 10^3 \Omega \text{ m}$
- (c) $2.77 \times 10^3 \Omega \text{ m}$
- (d) $0.0277 \times 10^3 \Omega \text{ m}$

Ans : (c)

60. The compound gallium arsenide has an intrinsic conductivity of $10^{-6} \text{ ohm}^{-1} \text{ m}^{-1}$ at 20°C . How many electrons have jumped the forbidden energy gap? [Given: $\mu_e = 0.88 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_h = 0.04 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$]

- (a) $6.79 \times 10^3 \text{ m}^{-3}$
- (b) $0.679 \times 10^3 \text{ m}^{-3}$
- (c) $67.9 \times 10^3 \text{ m}^{-3}$
- (d) $0.0679 \times 10^3 \text{ m}^{-3}$

Ans : (a)